Data Collection Tool: Wayne State Open Spine Consortium

Protocol: Creation of a Secure Multimodal Clinical and Radiographic Spine Surgery Dataset for Case-Control Research and Locally Hosted AI Development

1. Administrative & Linkage (Secure REDCap Instrument)

Note: This instrument contains the PHI master key and will be separated from the final dataset via REDCap Data Access Groups.

Variable Name	**Field Label (Question)**	**Field Type**	**Choices / Calculations / Validation**
study_id	Unique Study Identifier	Text Box	*Validation: WSS-####*
mrn	Medical Record Number	Text Box	*Validation: Integer*
date_of_surgery	Date of Index Surgery	Text Box	*Validation: YYYY-MM-DD*
date_of_birth	Patient Date of Birth	Text Box	*Validation: YYYY-MM-DD*
age_at_surgery	Calculated Age at Surgery	Calculated	datediff([date_of_birth], [date_of_surgery], 'y')

2. Preoperative Demographics & Clinical Status

Variable Name	Field Label (Question)	Field Type	Choices / Calculations / Validation
sex	Biological Sex	Radio	1, Male | 2, Female
bmi	Body Mass Index	Text Box	*Validation: Number (10.0 - 60.0)*
asa_class	ASA Physical Status	Dropdown	1, I | 2, II | 3, III | 4, IV | 5, V
smoking_status	Smoking Status	Radio	0, Never | 1, Former | 2, Current
primary_dx	Primary Surgical Diagnosis	Dropdown	1, Spondylolisthesis | 2, Stenosis | 3, Herniation | 4, Deformity | 5, Trauma/Tumor

3. Radiographic & Deep Learning Labels

Note: These variables establish the ground truth for the TotalSpineSeg and SPINEPS computational pipelines.

Variable Name	Field Label (Question)	Field Type	Choices / Calculations / Validation
lstv_present	Presence of LSTV on imaging?	Radio	0, No | 1, Yes

Variable Name	Field Label (Question)	Field Type	Choices / Calculations / Validation
castellvi_type	Castellvi Classification (if LSTV)	Dropdown	1, I (a/b) | 2, II (a/b) | 3, III (a/b) | 4, IV
nifti_file_id	De-identified NIfTI File Link	Text Box	*File nomenclature (e.g., sub-0001_T2w.nii.gz)*
mri_slice_thickness	MRI Slice Thickness (mm)	Text Box	*Validation: Number*

4. Intraoperative Surgical Details

Variable Name	Field Label (Question)	Field Type	Choices / Calculations / Validation
procedure_type	Primary Procedure	Dropdown	1, Decompression Only | 2, TLIF/PLIF | 3, ALIF/LLIF | 4, PSF
levels_fused	Number of Levels Fused	Text Box	*Validation: Integer (0 - 15)*
upper_instrumented	Upper Instrumented Vertebra (UIV)	Dropdown	1, T10 | 2, L1 | 3, L4 (etc.)

Variable Name	Field Label (Question)	Field Type	Choices / Calculations / Validation
lower_instrumented	Lower Instrumented Vertebra (LIV)	Dropdown	1, L5 | 2, S1 | 3, Pelvis/Ilium
ebl	Estimated Blood Loss (mL)	Text Box	*Validation: Integer*
op_time	Operative Time (minutes)	Text Box	*Validation: Integer*

5. Postoperative Clinical Outcomes

Variable Name	Field Label (Question)	Field Type	Choices / Calculations / Validation
length_of_stay	Length of Stay (Days)	Text Box	*Validation: Integer*
discharge_disp	Discharge Disposition	Dropdown	1, Home | 2, Rehab/SNF | 3, LTAC | 4, Expired
dural_tear	Incidental Dural Tear?	Radio	0, No | 1, Yes
ssi_30d	Surgical Site Infection (30-day)?	Radio	0, No | 1, Yes (Superficial) | 2, Yes (Deep)

Variable Name	Field Label (Question)	Field Type	Choices / Calculations / Validation
readmission_30d	30-Day Readmission?	Radio	0, No | 1, Yes
reoperation_90d	90-Day Reoperation?	Radio	0, No | 1, Yes